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Sustainable Investing and Market Governance

Contents

1	Big picture	3
2	Model and economic objects (key objects)	4
2.1	Timeline and agents	4
2.2	Firm payoff, externality, and effort	4
2.3	Why these objects matter	4
2.4	Trading environment	5
2.5	Preferences of the informed investor	5
2.6	Shareholders' contracting problem	6
2.7	Assumptions doing the main work	6
3	Benchmark with only financial investors	6
3.1	Benchmark trading and pricing	6
3.2	Likelihood ratios and effort informativeness	7
3.3	Optimal benchmark contract	7
4	Agency cost of sustainable investing (all equations + interpretation)	8
4.1	Negative externality: sustainable investors sometimes refuse to buy good financial news	8
4.2	Positive externality: sustainable investors may buy bad financial news	9
4.3	The ES–governance relationship	10
4.4	No greenium, but volatility effects	10
5	What assumptions are doing the work	11
6	Extensions and robustness	11
6.1	Public news about the externality and ES-linked compensation	11
6.2	Shareholder investment to reduce the externality	12
6.3	Managerial social effort	12

6.4	Precision of private information about externalities	12
6.5	Correlation between financial payoff and externality	13
6.6	Prevalence of sustainable investors	13
7	Figures	13
7.1	Figure 1: Model timeline	14
7.2	Figure 2: Trading strategy with a negative externality	14
7.3	Figure 3: Equilibrium trading strategy as γ rises	14
7.4	Figure 4: Equilibrium prices	15
7.5	Figure 5: Maximum likelihood ratio	15
7.6	Figure 6: Cost of incentive provision	15
8	Empirical implications and discussion	16
8.1	Empirical implications the paper wants you to test	16
8.2	Broader discussion points	17
9	Short summary	17
10	Conclusion	17

1 Big picture

- **Question.** How does sustainable investing affect the *governance role* of financial markets, not just the cost of capital or firms' externalities?
- **Core setting.** A manager can exert costly effort to improve the firm's financial payoff. An informed investor learns both the firm's financial payoff and its externality before trading. The investor may care about the externality with intensity γ , so she does not trade purely on financial information.
- **Main mechanism.** If informed investors care about externalities, they trade less aggressively on financial fundamentals. That makes prices less informative about managerial effort. Since shareholders use prices to incentivize the manager, incentive provision becomes more expensive.
- **Main governance result.** Sustainable investing can weaken market monitoring: stronger investor concern about externalities lowers price informativeness, raises agency costs, and can make it privately optimal for shareholders *not* to induce managerial effort.
- **Why this is novel.** The paper identifies a *market-governance channel* of sustainable investing. The key real effect does not come from a higher cost of capital; it comes from a noisier signal for compensation design.
- **Negative versus positive screening.**
 - With **negative externalities** ($\eta < 0$), sustainable investors sometimes refuse to buy even when the firm has high financial payoff. That is negative screening.
 - With **positive externalities** ($\eta > 0$), sustainable investors may buy even when the firm has low financial payoff. That is positive screening.
- **Common implication across both cases.** Both screening patterns reduce how strongly order flow responds to financial fundamentals. So both weaken governance.
- **ES-governance link.** The direction of the ES–governance relation depends on the sign of the externality.
 - If the firm imposes a negative externality, better ES quality means *more* informed trading on financial information, so governance is *better*.
 - If the firm generates a positive externality, better ES quality means *less* informed trading on financial information, so governance is *worse*.
- **Asset-pricing implication.** The model implies *no greenium* in expected returns because risk-neutral market makers price only expected financial payoffs. But sustainable investing still changes *volatility* and *real outcomes*.
- **Real-effects implication.** Because sustainable investing changes governance, it can induce financially motivated shareholders or managers to change externalities. This can be socially beneficial for *negative* externalities, but potentially harmful for *positive* externalities.
- **Contracting implication.** ES-linked compensation can be optimal even when shareholders only care about financial payoffs and even when managers cannot directly affect ES outcomes. The reason is informational: ES signals help decode what prices mean about effort.
- **Economic takeaway.** Sustainable investing is not only about “pricing sustainability.” It also changes *how informative prices are*, and therefore how well markets discipline managers.

2 Model and economic objects (key objects)

2.1 Timeline and agents

- There are three dates, $t \in \{0, 1, 2\}$, and all agents are risk-neutral.
- **At $t = 0$:** the firm's initial controlling shareholders design a compensation contract for the manager. The manager can exert costly *financial effort* $e_F \in \{0, 1\}$.
- **At $t = 1$:** an informed investor privately learns both the realized financial payoff and the realized externality, and then trades in a discrete Kyle-type market.
- **At $t = 2$:** the financial payoff and the externality are realized for society and the firm's owners.

2.2 Firm payoff, externality, and effort

- The firm has N shares outstanding and generates financial payoff per share

$$F \in \{0, 1\}.$$

- The firm also generates an externality

$$E \in \{0, \eta\},$$

where $\eta < 0$ means a negative externality and $\eta > 0$ means a positive externality.

- If the manager exerts financial effort $e_F = 1$, the probability of high financial payoff is $p_F \in (0, 1)$.
- If the manager shirks ($e_F = 0$), the probability of high financial payoff falls to

$$p_F - \Delta_F, \quad 0 < \Delta_F < p_F.$$

- Shirking gives the manager a private benefit $B_F > 0$.
- The probability of a low externality $E = 0$ is

$$p_E \in (0, 1),$$

so the probability of the high externality $E = \eta$ is $1 - p_E$.

- In the baseline model, the financial payoff F and the externality E are independent, and p_E is exogenous.

2.3 Why these objects matter

- Δ_F measures how much managerial effort matters for firm value.
- B_F measures how severe the agency problem is.
- p_E measures how often the firm avoids the externality. For negative externalities, higher p_E means better ES quality. For positive externalities, higher p_E means worse ES quality because the positive externality occurs less often.
- η measures the *magnitude* of the externality, while $1 - p_E$ measures the *frequency*. The paper emphasizes that both work in the same direction.

2.4 Trading environment

- The informed investor can submit an order

$$x \in \{0, 1\},$$

so she either buys one share or abstains.

- Noise traders have random demand

$$z \in \mathbb{N}_0 = \{0, 1, 2, \dots\},$$

with geometric distribution

$$\mathbb{P}(z) = (1 - \lambda)^z \lambda, \quad \lambda \in (0, 1).$$

- Aggregate order flow is

$$q = x + z.$$

- Market makers are competitive and set the stock price to expected *financial payoff* conditional on observable order flow:

$$P(q) = \mathbb{E}[F \mid q].$$

Interpretation.

- The market price is not a direct reflection of ES value. It is a signal extraction device for expected financial payoff.
- That is crucial: if sustainable investors trade on ES concerns, order flow becomes a noisier signal of financial fundamentals.
- The paper is therefore about *price informativeness*, not directly about discount rates.

2.5 Preferences of the informed investor

The informed investor cares about both the financial payoff and the externality. If she owns one share, her gross utility is

$$x(F + \gamma E), \tag{1}$$

where $\gamma \geq 0$ is the intensity of social concerns.

Her net utility is

$$x(F + \gamma E - \mathbb{E}[P]). \tag{2}$$

Interpretation.

- When $\gamma = 0$, the informed investor is purely financial.
- When $\eta < 0$, a larger γ makes the investor less willing to buy firms with bad externalities.
- When $\eta > 0$, a larger γ makes the investor willing to buy firms with good externalities even if financial payoff is weak.
- This wedge between *financial information* and *trading behavior* is the source of the whole mechanism.

2.6 Shareholders' contracting problem

- Initial controlling shareholders are purely financial investors. They care only about the firm's expected financial payoff net of compensation costs.
- The manager's compensation is constrained by limited liability,

$$W \geq 0,$$

and can depend only on the stock price, equivalently on order flow q .

- The manager's outside option is normalized to zero.

Why this assumption matters.

- The paper wants to isolate the governance role of prices.
- If shareholders also intrinsically valued ES outcomes, there would be a separate *voice* channel. The baseline strips that out and isolates the effect coming from market monitoring.

2.7 Assumptions doing the main work

- **Risk-neutral agents and competitive market makers:** this shuts down a cost-of-capital channel and lets the paper isolate the information channel.
- **Price depends on expected financial payoff only:** the market price is a financial signal. That is why trading on ES concerns dilutes the signal.
- **Managerial effort is socially efficient:**

$$N\Delta_F > B_F.$$

This means it is socially desirable to induce effort; if effort is not induced, that is a real inefficiency.

- **Baseline independence of F and E :** the paper makes the cleanest possible point first. Later it shows the mechanism survives correlation.

3 Benchmark with only financial investors

The benchmark sets $\gamma = 0$. Then the informed investor cares only about F and the model collapses to a standard market-monitoring environment.

3.1 Benchmark trading and pricing

Proposition 1. If the manager exerts effort, there is a unique equilibrium in which the informed investor buys one share if and only if $F = 1$.

The benchmark pricing rule is

$$P_0(q) = \begin{cases} 0, & q = 0, \\ \frac{p_F}{p_F + (1 - p_F)(1 - \lambda)}, & q > 0. \end{cases} \quad (3)$$

Interpretation.

- If $q = 0$, the market infers that no informed buy order arrived, so the firm must have low financial payoff.
- If $q > 0$, the market sees either informed buying because $F = 1$ or noise buying. The price is therefore strictly between 0 and 1.
- Positive order flow is good news, but not perfectly revealing news.

3.2 Likelihood ratios and effort informativeness

Define the likelihood ratio of order flow $q = k$ under effort versus shirking as

$$\phi_0(k) = \frac{\mathbb{P}(q = k \mid e_F = 1)}{\mathbb{P}(q = k \mid e_F = 0)}. \quad (4)$$

The paper shows

$$\phi_0(k) = \begin{cases} \frac{1 - p_F}{1 - p_F + \Delta_F}, & k = 0, \\ \frac{p_F \lambda + (1 - \lambda)}{(p_F - \Delta_F) \lambda + (1 - \lambda)}, & k > 0. \end{cases} \quad (5)$$

So the maximum likelihood ratio is attained when $q > 0$:

$$\phi_0^* = \frac{p_F \lambda + (1 - \lambda)}{(p_F - \Delta_F) \lambda + (1 - \lambda)}. \quad (6)$$

Interpretation.

- The stock price is informative about effort because effort changes the probability of positive order flow.
- Positive order flow is the most informative state, so the optimal contract will reward the manager there.
- Higher noise-trader precision (higher λ) raises informativeness because order flow is then less contaminated by noise.

3.3 Optimal benchmark contract

Corollary 1. The optimal incentive-compatible contract is

$$W_0^*(q) = \begin{cases} 0, & q = 0, \\ \frac{B_F}{\Delta_F \lambda}, & q > 0. \end{cases} \quad (7)$$

The expected cost of providing incentives is

$$C_0 = \frac{1}{1 - 1/\phi_0^*} B_F. \quad (8)$$

Effort is induced if and only if

$$N\Delta_F \geq C_0. \quad (9)$$

Interpretation.

- The manager gets paid only when the market sees positive order flow, that is, when the signal about effort is strongest.
- The better the signal ϕ_0^* , the cheaper it is to incentivize effort.
- This benchmark gives the clean object the paper will later destroy: *price informativeness about effort*.

4 Agency cost of sustainable investing (all equations + interpretation)

4.1 Negative externality: sustainable investors sometimes refuse to buy good financial news

Now suppose the firm imposes a negative externality, so $\eta < 0$.

The informed investor's equilibrium behavior is simple in three states:

- if $F = 0$, she never buys regardless of E ;
- if $F = 1$ and $E = 0$, she always buys;
- if $F = 1$ and $E = \eta < 0$, she buys only with probability $a \in [0, 1]$.

Lemma 2. There are thresholds $\underline{\gamma} < \bar{\gamma}$ such that:

- if $\gamma \leq \underline{\gamma}$, then $a^* = 1$;
- if $\underline{\gamma} < \gamma < \bar{\gamma}$, then $a^* \in (0, 1)$ and decreases with γ ;
- if $\gamma \geq \bar{\gamma}$, then $a^* = 0$.

Define the equilibrium probability of informed buying conditional on high financial payoff as

$$\tau_\gamma^* = p_E + (1 - p_E)a^*. \quad (10)$$

Interpretation.

- τ_γ^* is the key sufficient statistic for how much the informed investor still trades on financial information.
- When social concerns are weak, $\tau_\gamma^* = 1$ and the model collapses to the benchmark.
- When social concerns are strong, the investor buys only when the high payoff is accompanied by a low externality, so $\tau_\gamma^* = p_E$.

The maximum likelihood ratio becomes

$$\phi_\gamma^* = \frac{\lambda p_F \tau_\gamma^* + (1 - \lambda)}{\lambda(p_F - \Delta_F) \tau_\gamma^* + (1 - \lambda)}. \quad (11)$$

Proposition 2. For sufficiently strong social concerns,

$$\phi_\gamma^* < \phi_0^*. \quad (12)$$

Interpretation.

- The price becomes less informative because the informed investor no longer buys every time she sees good financial news.
- Order flow now mixes two things: financial information and social screening.
- The market sees less of the manager's effect on F , so the price is a worse performance signal.

The optimal contract becomes

$$W_\gamma^*(q) = \begin{cases} 0, & q = 0, \\ \frac{B_F}{\Delta_F \lambda \tau_\gamma^*}, & q > 0, \end{cases} \quad (13)$$

with expected incentive cost

$$C_\gamma = \frac{1}{1 - 1/\phi_\gamma^*} B_F. \quad (14)$$

Corollary 2. For sufficiently strong social concerns,

$$C_\gamma > C_0. \quad (15)$$

If $C_\gamma > N\Delta_F$, shareholders optimally stop providing incentives and the manager shirks.

Interpretation.

- Sustainable investing raises the cost of governance because the manager must be rewarded more aggressively when the signal is worse.
- This can create a real output loss: socially efficient effort is no longer induced.
- That loss is the paper's **agency cost of sustainable investing**.

4.2 Positive externality: sustainable investors may buy bad financial news

Now suppose the firm generates a positive externality, so $\eta > 0$.

Then the distortion changes sign:

- if $F = 1$, the investor buys regardless of E ;
- if $F = 0$ and $E = 0$, she abstains;
- if $F = 0$ and $E = \eta > 0$, she may still buy because the positive externality gives utility.

There is a threshold $\bar{\gamma}$ such that for sufficiently strong social concerns the investor buys even when $F = 0$ and $E = \eta$.

Proposition 3. The governance implication is the same as in the negative-externality case: for sufficiently strong social concerns, incentive costs rise relative to the benchmark, and if they rise enough, shareholders stop inducing effort.

Interpretation.

- With negative externalities, the distortion is *underreaction to good financial news*.
- With positive externalities, the distortion is *overreaction to good ES news even when financial news is bad*.
- These are different trading patterns, but they have the same governance consequence: prices are less informative about the manager's financial effort.

4.3 The ES–governance relationship

This is the paper's clean comparative-static result.

Proposition 4. When social concerns are sufficiently strong, effort informativeness ϕ_γ^* increases with p_E . The implied relation between ES quality and governance depends on the sign of the externality.

Negative externality ($\eta < 0$).

- Higher p_E means the firm is less likely to impose the bad externality.
- So the informed investor trades more often on financial information.
- Therefore better ES quality implies **better governance**.

Positive externality ($\eta > 0$).

- Higher p_E means the firm is less likely to generate the good externality.
- Since ϕ_γ^* increases with p_E , governance is stronger when positive externalities are *less* likely.
- Therefore better ES quality implies **worse governance**.

Key intuition. The price is most informative when the informed investor trades on financial information alone. Anything that makes the externality matter *less often* sharpens governance. For negative externalities, that corresponds to better ES quality. For positive externalities, that corresponds to worse ES quality.

4.4 No greenium, but volatility effects

The paper deliberately shuts down a cost-of-capital channel. Since market makers are risk-neutral and price only expected financial payoffs,

$$\mathbb{E}[F - P_\gamma(q)] = 0, \tag{16}$$

so firms with different ES qualities have identical expected returns.

Interpretation.

- No greenium does *not* mean sustainable investing is irrelevant.
- Sustainable investing still changes trading behavior, price informativeness, compensation costs, effort, and externalities.

Proposition 5. Stock-price volatility increases with p_E . Hence:

- for negative externalities, better ES quality implies **higher volatility**;

- for positive externalities, better ES quality implies **lower volatility**.

Interpretation.

- In this model, higher volatility reflects more informed trading on financial fundamentals.
- So the volatility result mirrors the governance result.
- The paper’s point is subtle: sustainable investing can change higher moments of returns even when average expected returns stay unchanged.

5 What assumptions are doing the work

- **Investors care about externalities, but market makers price financial payoff.** This wedge is the basic source of less informative prices.
- **Compensation depends on price.** If shareholders could contract directly on managerial effort, the informational role of prices would not matter.
- **Initial shareholders are purely financial.** This ensures the real effects come from governance rather than shareholders directly wanting better ES outcomes.
- **Active informed capital is scarce.** The later extension shows the mechanism depends on the prevalence of sustainable investors among *informed* investors, not among passive holders.
- **The mechanism does not require a greenium.** That is one of the paper’s strongest conceptual points: sustainable investing can matter even with no expected-return differential.

6 Extensions and robustness

6.1 Public news about the externality and ES-linked compensation

Suppose that after trading there is public news $\sigma \in \{L, H\}$ about the externality, with $\sigma = L$ indicating a lower expected externality.

Proposition 6. For sufficiently strong social concerns, the optimal contract conditions on both order flow q and the public ES signal σ .

Corollary 3. Adding the ES signal lowers incentive costs and can switch the firm from not inducing effort to inducing effort.

Interpretation.

- ES-linked compensation is valuable here because it helps shareholders interpret what order flow means about managerial effort.
- The contract is not necessarily trying to make the manager “care about ESG.” It is using ES information to recover a cleaner effort signal.
- This gives the paper a distinct explanation for ES-linked pay: *information extraction*, not necessarily social preference alignment.

Important asymmetry.

- If the firm imposes a negative externality, the manager is rewarded when $q > 0$ and the ES signal is favorable.
- If the firm generates a positive externality, the logic can imply rewarding low externality states or, equivalently, punishing high externality states, because those are the states where prices are more informative about financial effort.

6.2 Shareholder investment to reduce the externality

Suppose shareholders can pay cost $c > 0$ to raise the probability of a low externality from p_E to $p_E + \Delta_E$.

Proposition 7. For sufficiently strong social concerns, such investment lowers incentive costs. There exists a threshold $\bar{c} > 0$ such that shareholders invest if and only if $c \leq \bar{c}$.

Interpretation.

- Financially motivated shareholders may improve ES quality *because* doing so improves governance.
- For negative externalities, this gives a “doing well by doing good” result: reducing the bad externality sharpens price informativeness and lowers compensation costs.
- For positive externalities, the exact same logic can perversely induce shareholders to *reduce* positive externalities, because a smaller externality makes prices more informative.

Key takeaway. The mechanism is socially attractive for bad externalities and socially perverse for good externalities.

6.3 Managerial social effort

Now let the manager also choose social effort $e_E \in \{0, 1\}$, which improves ES quality. The manager gets private benefit $B_E > 0$ from shirking on that dimension.

Proposition 8. If the agency problem for financial effort is more severe, any contract that induces financial effort also induces social effort.

Proposition 9. If the agency problem for financial effort is less severe, the optimal contract does not induce social effort.

Interpretation.

- This is a multitasking extension, but the paper gets a result very different from the classic Holmström–Milgrom tension.
- For negative externalities, financial effort and social effort can be *complements*: social effort improves the informativeness of prices about financial effort.
- But whether that complementarity is exploited depends on whether the bonus needed to induce financial effort is already powerful enough to induce social effort too.

6.4 Precision of private information about externalities

The paper relaxes perfect observability of E . The informed investor now gets a noisy signal about the externality, with precision parameter $\psi \in [0, 1]$.

Proposition 10. Increasing signal precision can either improve or worsen effort informativeness.

Interpretation.

- If the high externality is very likely, better ES information can help the investor sometimes realize the firm is actually not so bad, so she trades more on financial information.
- If the high externality is initially unlikely, better ES information can create more occasions when the investor deviates from financial trading, making governance worse.
- So better ESG analytics do not automatically improve governance.

6.5 Correlation between financial payoff and externality

Proposition 11. The governance channel survives as long as the financial payoff and the externality are not perfectly correlated in a way that eliminates the distortion state.

Corollary 4.

- With negative externalities, *positive* correlation between financial payoff and bad externality amplifies the governance cost.
- With positive externalities, *negative* correlation between financial payoff and good externality amplifies the governance cost.

Interpretation.

- The stronger the correlation that makes the distortion state more common, the stronger the market-governance channel.
- This gives the paper a clean source of cross-industry heterogeneity.

6.6 Prevalence of sustainable investors

The paper finally allows the informed trader to be a sustainable investor with probability n_S and a purely financial investor with probability $1 - n_S$.

Proposition 12. When sustainable investors' social concerns are sufficiently strong, incentive costs increase strictly with n_S .

Interpretation.

- The relevant object is not total sustainable AUM in the economy.
- The relevant object is the fraction of *active, information-producing traders* who are sustainable investors.
- Passive green capital does not drive this mechanism; active informed green capital does.

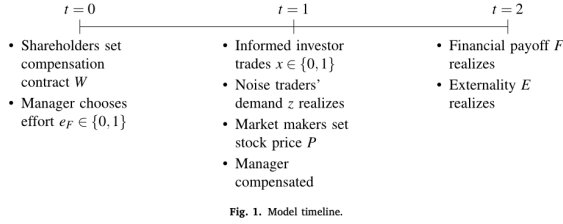
7 Figures

There are no empirical tables in this paper. It is a theory paper. The important visuals are the figures that map the mechanism from trading to prices to contracting costs.

7.1 Figure 1: Model timeline

Read: This figure is the whole architecture of the paper.

- At $t = 0$, shareholders write a stock-price-based contract and the manager chooses effort.
- At $t = 1$, the informed investor learns (F, E) and trades.
- At $t = 2$, payoff and externality realize.
- The figure makes clear that prices are a *signal* feeding into compensation, not just an asset-pricing object.



		Externality E	
		$E = 0$ w.p. p_E	$E = \eta < 0$ w.p. $1 - p_E$
Financial payoff F	$F = 0$ w.p. $1 - p_F$	$x = 0$	$x = 0$
	$F = 1$ w.p. p_F	$x = 1$	$x = 1$ w.p. a

Fig. 2. Informed investor's trading strategy with a negative externality. This figure illustrates the informed investor's equilibrium trading behavior conditional on observing the financial payoff (F) and externality (E) realizations. Variable $a \in [0, 1]$ denotes the probability of buying upon observing $F = 1$ and $E = \eta$.

7.2 Figure 2: Trading strategy with a negative externality

Read: This figure gives the negative-screening logic in one picture.

- Buy for sure when $F = 1$ and $E = 0$.
- Do not buy when $F = 0$.
- When $F = 1$ and $E = \eta < 0$, buy only with probability a .
- This is the state where sustainable investing destroys price informativeness: the investor has good financial news but partially refuses to act on it.

7.3 Figure 3: Equilibrium trading strategy as γ rises

Read: The left panel shows a^* falling with γ ; the right panel shows the effective informed-trading intensity τ_γ^* falling too.

- Weak social concerns: benchmark trading.

- Intermediate concerns: mixed strategy.
- Strong concerns: no buying in the high-financial-payoff/high-negative-externality state.

Why it matters. This figure is the direct bridge from preferences to governance: stronger ES concerns $\Rightarrow$ less trade on financial information.

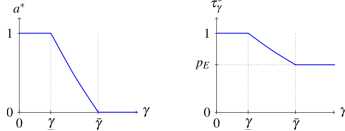


Fig. 3. Equilibrium trading strategy. This figure shows how the informed investor's trading strategy varies with the intensity of her social concerns (γ). The left panel plots the probability of buying when both the financial payoff and the externality are high (a^*). The right panel plots the overall probability of informed buying given a high financial payoff (τ_γ^*).

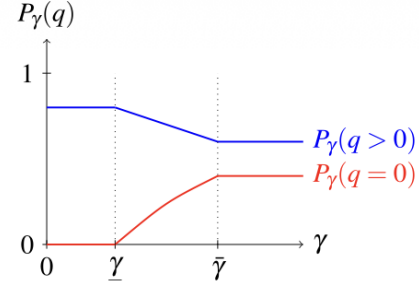


Fig. 4. Equilibrium prices. This figure shows how equilibrium prices vary with the intensity of the informed investor's social concerns (γ). The blue line represents the price when aggregate order flow is high ($q > 0$), and the red line represents the price when order flow is low ($q = 0$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

7.4 Figure 4: Equilibrium prices

Read: The figure shows that high and low order-flow prices move closer together as γ rises.

- The positive-order-flow price falls.
- The zero-order-flow price rises.
- The market becomes less responsive to order flow because order flow says less about F .

Why it matters. Smaller price separation means a weaker signal for the compensation contract.

7.5 Figure 5: Maximum likelihood ratio

Read: This is the cleanest governance figure in the paper.

- The maximum likelihood ratio ϕ_γ^* is flat at the benchmark for low γ .
- It declines once social concerns become strong enough to distort trading.

Why it matters. This figure literally shows market monitoring deteriorating as sustainable-investor concerns strengthen.

7.6 Figure 6: Cost of incentive provision

Read: This figure converts the information result into a real contracting cost.

- Incentive cost is flat at low γ .

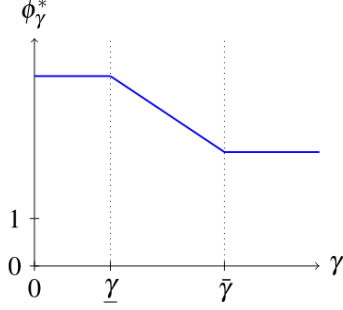


Fig. 5. Maximum likelihood ratio. This figure shows how the maximum likelihood ratio ϕ_γ^* varies with the informed investor's social concerns.

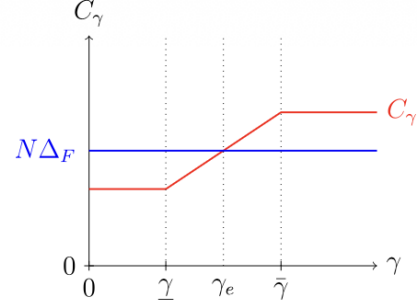


Fig. 6. Cost of incentive provision. This figure plots the expected cost of providing incentives (C_γ) as a function of the intensity of the investor's social concerns (γ). The threshold γ_e represents the level of social concerns above which managerial effort is no longer induced.

- It rises with γ once the trading distortion appears.
- There is a threshold above which the firm no longer induces managerial effort.

Why it matters. This is the main welfare figure of the paper: less informative prices become higher agency costs and possibly lower firm performance.

8 Empirical implications and discussion

8.1 Empirical implications the paper wants you to test

- **Growth of active sustainable investing.** As active sustainable investors become more prevalent, prices should become less informative, governance weaker, and stock-based incentives less common.
- **Among firms that still use stock-based incentives,** pay-for-performance sensitivity and expected compensation should rise because the same incentive now requires a bigger bonus.
- **ES-linked compensation** should be more common when public ES signals are more useful for decoding price signals.
- **Negative screening predicts:** firms with worse ES quality should have weaker governance and lower volatility.
- **Positive screening predicts:** firms with better ES quality should have weaker governance and lower volatility.
- **Net-zero commitments** should be attractive because they remove externality-driven distortions and maximize price informativeness. The paper even suggests this can explain why net-zero, rather than carbon-negative, is a focal point.
- **Frequency and magnitude complementarity.** The mechanism should be strongest for firms whose externalities are both frequent and large.

8.2 Broader discussion points

- **Consequentialist versus warm-glow preferences.** The model is written with direct utility from externalities, but the same trading patterns can arise from strategic consequentialist investing: an investor may avoid bad firms to induce them to improve.
- **Welfare asymmetry.** Sustainable investing is more likely to generate socially useful changes when firms impose negative externalities than when they generate positive externalities.
- **Investor sorting.** Sustainable informed investors may specialize in greener firms. That can amplify governance differences across firms by reallocating scarce informed capital.
- **What counts as governance?** If governance means “maximizing financial value,” the ES–governance relation can be positive or negative depending on the sign of the externality. If governance includes social objectives, interpretation changes.
- **Carbon tax.** The paper argues that carbon taxation can improve both social and financial efficiency: it makes it privately valuable for shareholders to reduce negative externalities, which in turn improves market monitoring.

9 Short summary

- Sustainable investing affects firms not only through the cost of capital, but also through the *informativeness of prices*.
- When informed investors care about externalities, they trade less purely on financial fundamentals. That weakens market monitoring and raises the cost of managerial incentives.
- The resulting relationship between ES and governance depends on whether sustainable investing is based on negative screening or positive screening.
- The model predicts no greenium in expected returns, but strong effects on volatility, compensation design, effort provision, and firms’ incentives to change externalities.
- ES-linked compensation can be optimal for a purely financial reason: it helps interpret prices, not necessarily because shareholders intrinsically care about ESG.

10 Conclusion

- **Core conclusion.** Sustainable investing can weaken the governance function of financial markets by making prices less informative about managerial effort.
- **Economic intuition.** The market disciplines managers only when informed trading reveals financial fundamentals. Once informed investors also trade on externalities, order flow becomes a noisier governance signal.
- **Takeaway.** Sustainable investing can create real effects even without changing expected returns. The critical object is not just who owns capital, but who produces information and how their objectives shape trading.